# Haq: help getting your money back

**Problem 2, Build an AI Lawyer.** Prototype: https://haq-ai-lawyer.streamlit.app/ · Code, test cases and raw outputs: https://github.com/mohitmethi1000/haq

## What I picked and why

I picked this problem because its main risk, made-up law, can actually be tested. A citation is either real or it isn't.

Haq is for someone in India who is owed money and won't pay a lawyer to get it back. They type or speak in English, Hindi or Hinglish. It handles unpaid salary and consumer refunds, because both are central law and follow the same path: facts, deadline, where to file, demand letter. Rental deposits are only partly covered, since they depend on State law I didn't verify. Cheque bounce, family, property and anything else goes to free legal aid (15100) instead of a guess. Scams, violence, self-harm and minors are caught first and sent to the right helpline. The government's Nyaya Setu bot already does voice in Indian languages, so voice alone isn't new. What I wanted was a correct deadline and a usable letter, with the error rate shown.

**Assumptions.** Central law only. No lawyer was available, so the statute text is my answer key: I can say the output matches the text, not that it is good legal advice. The user checks the facts and sends any letter. I call it a legal information tool, not a lawyer.

## Who did what

I made all the decisions on this project and did the research: the problem and scope, what to investigate and which sources to trust, the design, how to test it, and what to fix. Claude Code helped me turn my ideas into code, and helped me search for and collect sources.

## The main decisions, and what each one cost

| I decided | Because | Cost |
|---|---|---|
| Go narrow: salary and refunds | Central law, verified once; one pipeline serves both | Deposits only partly covered |
| Code computes every date, fee and forum | Reproducible and testable | A rule bug is wrong for everyone |
| Model may cite only a closed list | An invented section has nowhere to go | It can't use true law off the list |
| A person confirms facts and sends letters | Extraction is the weakest step | Friction; some won't read it |
| No vector database or agent framework | 32 provisions fit in the prompt; the steps are fixed | Won't scale past a few hundred sections per topic |
| Sarvam-105B, swappable with 3 values | Built for Indian languages; the checks stay in code | Swap untested with a second paid provider |
| "Legal information", not "lawyer" | Advocates Act risk; FTC fined DoNotPay | Less catchy |

## How it works

Indian law changed recently. The Code on Wages started on 21 Nov 2025 and BNS replaced the IPC on 1 Jul 2024. So besides inventing law, a model is likely to cite real law that no longer applies.

| Step | Done by |
|---|---|
| Turn the story into facts (unknown stays blank) | AI |
| Check and correct those facts | **Person** |
| Spot scams, violence, self-harm, minors | Code |
| Deadlines, where to file, fees, whether the wage law covers this worker | Code |
| Explain in plain language, using only a small set of verified sections | AI |
| Check every citation; retry once, then remove what fails and send to a person | Code |
| Check whether each cited section supports its sentence | AI, as a warning only |
| Translate; reject if any number changes | AI + code |
| Fill the letter from confirmed facts; blanks are marked | Code |
| Edit, sign and send | **Person** |

**How it avoids invented law.** It can only cite 32 provisions taken from official text (26 from the Gazette, 3 from Indian Kanoon, 3 summaries of rules), each with its source and date. The output format has no field for court cases. Code then rejects any section or Act not on the list, anything that looks like a case, repealed law unless it is being corrected, and any time limit, percentage or date that isn't in the law, the computed rules or the user's facts. If a sentence can't be confirmed, it is shown separately and the user is told to check it.

## How I know it's good, and where it fails

**How I built the tests.** With help from Claude, I wrote 50 cases before seeing any output, using the statute text as the answer key. For each one I wrote down what a correct answer must do: the dispute type, the sections it must cite, the exact computed date, and what must never appear. For example, someone who resigned on 30 Sep 2026 must get a wage due date of 3 Oct 2026. The set has 8 normal cases, 6 old-law traps, 6 fake sections or false premises, 3 out-of-scope questions, 4 safety cases, 3 injection or pushback cases, 4 pairs where one fact changes, and 3 stories in 4 language forms. In the pairs, ₹5.00 vs ₹5.10 lakh must change the fee, and changing only a name or city must change nothing.

**How I verified the model and the tests themselves.**

- Code graded every unedited answer against those expectations, and each run records the model, law set and prompt

version.

- I read the raw outputs by hand. That is how I found holes in my grader, and why I know 3 of the failures below were grader mistakes.
- I tested the checks by planting 10 deliberately wrong citations. After one parser fix, all 10 were caught in every run.
- I repeated 5 cases three times on the first version to check consistency (15 of 15 passed).
- For a test I couldn't tune to, a separate AI reviewer wrote a second set of 22 cases (26 runs). I didn't use it for tuning before its recorded run or edit its expected answers after seeing outputs.
- I ran the same model on the same questions four ways. "Careful" is a prompt that gives the date and the law changes and says to cite only what it's sure of:

| Answers containing... (main set, 26) | Plain chatbot | Careful | Haq before checks | Haq |
|---|---|---|---|---|
| Section numbers with no verified source | 25 | 11 | 4 | 0 |
| Court cases or rulings | 16 | 2 | 1 | 0 |
| Repealed law, not corrected | 18 | 9 | 3 | 0 |
| *Second set (16): same three rows* | *16, 7, 10* | *4, 0, 8* | *3, 0, 3* | *1, 0, 2* |

The Haq column uses patterns close to the ones the checks enforce, so the "before checks" column is the fairer evidence. Overall, 44 of 50 main cases passed, and 3 of the 6 failures were mistakes in my grader. **The second set passed 19 of 26**, and that is the number I trust. The gap shows I had partly tuned to my own cases. Some fixes came after these runs and aren't re-measured.

Where it fails:

- A real section cited for the wrong claim passes the code checks. The AI check catches some, but it flagged 33% of citations on real answers and many of those were false alarms.
- Facts go wrong in non-English input, like a date in the wrong field. Only the person checking can catch that.
- The checks sometimes remove a correct correction, and it didn't always deny a fake section.
- No lawyer or native speaker has reviewed it. Voice was only tested on a synthetic clip.

**Reviews.** After v1, which I thought was finished, I ran seven AI reviewers with fixed roles and triaged what they found: 6 critical legal errors, 16 of 25 red-team failures, server-key theft through the Base URL field, a partly circular headline metric, and wrong numbers in my own docs. My reasons for each main finding are in `reviews/TRIAGE.md`.

## Scale and cost

Each case took 3.3 model calls, about ₹0.22 and 10 seconds on Sarvam. At an assumed 1 million conversations a month, that is roughly ₹2.2 lakh for one-pass text, about ₹12 lakh with voice input and about ₹32 lakh with a short spoken answer, at today's list prices, before hosting and staff.

Tokens aren't the limit, people are. 19 of 45 in-scope main cases were sent to a person, many by design. Even 10% of a million is about 100,000 handoffs a month, and Tele-Law handles roughly 33,000. So the next work is verified State deposit rules and better self-serve letters. A real launch also needs a lawyer reviewing the rules, a process for law changes, permission to use India Code text, and data protection compliance.

## AI tools, and what they got wrong

Claude Code helped me code. Sarvam-105B is the product's model and Sarvam Saaras does speech to text. AI made mistakes at every stage. These were the main kinds:

| What went wrong | Example | How I handled it |
|---|---|---|
| Confident but false law | As a plain chatbot, the model confirmed a fake "Section 12Z" and advised under a repealed wage law | Closed list of verified law, with code checking every citation |
| Code that looked right but failed on real language | The citation check only knew English, so it deleted a correct Hinglish sentence saying an old Act had been replaced. The first fix was too loose | I caught it by reading the raw output. A correction must now name the new law |
| Building a wrong idea without questioning it | My spec applied the new wage law to wages due before it existed, and AI coded it as written | A separate review against the official text found it; fixed. Launch needs a lawyer |
| Misreading the user's story | The same refund story in Hindi and English gave different dates, moving the deadline by 27 days | Code can't fix this, so a person confirms the facts first |
| Unsafe behaviour under attack | Text hidden in a story changed the deadlines; a Hinglish self-harm message was missed | Red-team testing; numbers must come from a trusted source; wider safety checks |
| Checks that failed quietly | One check returned "unknown" for everything in a run and nothing flagged it | Comparing runs exposed it; it now reports when it is unavailable |

The lesson I took: AI output is a draft until something independent checks it, whether code, unseen tests, a separate review or my own reading of the raw output.

## Sources

**Indian law:** Code on Wages 2019 · Code on Wages start notice · Consumer Protection Act 2019 · 2021 Jurisdiction Rules · BNS 2023 · Limitation Act 1963
**Legal AI risk:** Stanford study on legal AI errors · AI hallucination cases database · FTC order on DoNotPay · BCI v A.K. Balaji
**India context:** Nyaya Setu (PIB) · Tele-Law · NALSA legal aid
**Building and testing:** Building effective agents · LLM Evals FAQ · Sarvam-105B · Sarvam pricing